POSE ESTIMATION & LOOP CLOSURE

Wheel Odometry
Propagated Covariance

Node Pose Estimate

Mahalanobis Search
Previous Nodes

Loop Closure
Candidate Search

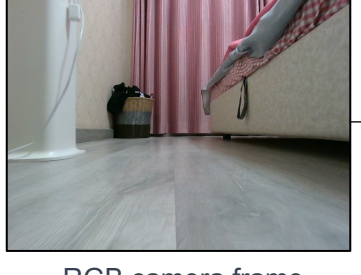
Candidate Resolution

No candidates	No loop edge
One candidate	Add loop edge
Several candidates	Bidirectional sequence; accept best or reject all

Activates Search

New edge

VISUAL TOPOLOGY CONSTRUCTION



RGB camera frame

RGB Stream
Camera Frames

DINOv2 Descriptor
$384 - D$

Node Extraction
λ_2 Monitoring

Topological Node
$vMF(\mu, \kappa)$

Topological Graph
Sequential + Loop Edges